Technical Indicators & Strategy Backtesting

Risk-adjusted comparison of three algorithmic trading strategies on MSFT using historical market data from Alpaca

Ticker: **MSFT** · Lookback: 5 years · Data: Alpaca Historical Market Data API (split & dividend adjusted) · Period: 2021-06-25 to 2026-06-29 · 1,257 daily bars · Generated: 2026-07-01 15:45 UTC

Initial capital **$100,000** · Long-only, no leverage, no short selling · Signals on daily close, executed at next open · 1 bp slippage per trade · Risk-free rate: 0.0% p.a.

1. Data

All backtests use daily OHLCV (Open-High-Low-Close-Volume) data for **MSFT** downloaded from **Alpaca Historical Market Data API (split & dividend adjusted)**. The Alpaca SDK fetches bars adjusted for splits and dividends (adjustment='all'), ensuring that historical returns are not distorted by corporate actions. Data is stored locally in Parquet format for fast re-runs without additional API calls.

Metric	Value
Start price	$254.39
End price	$368.36
Price return (full period)	44.8%
Annualised volatility	26.9%
Daily return skewness	0.032
Daily return excess kurtosis	3.573
Largest single-day gain	9.7%
Largest single-day loss	-10.1%
Avg daily volume	601,250 shares

The return distribution is approximately symmetric with excess kurtosis of 3.57 (positive kurtosis = fat tails relative to a normal distribution, meaning extreme days occur more frequently than a Gaussian model predicts). These distributional properties matter for the Sortino ratio, which penalises fat left tails more heavily than a standard Sharpe ratio.

2. Technical Indicators Implemented (11 total)

All 11 indicators are implemented from scratch in pure pandas/numpy — no black-box TA library dependency — so every calculation is mathematically transparent and auditable. The project requirement is a minimum of 6 indicators spanning at least two categories; this implementation covers **all four required categories**: Trend, Momentum, Volatility, and Volume.

Category	Indicator	Parameters	Strategy Usage
Trend	Simple Moving Average (SMA)	windows: 20, 50, 200	Trend Following — SMA-50 regime filter
Trend	Exponential Moving Average (EMA)	windows: 12, 26, 50, 200	Custom Triple Confirmation — EMA-50/200 golden cross
Trend	MACD & Signal Line	fast=12, slow=26, sig=9	Trend Following — MACD crossover entry/exit
Trend	Avg Directional Index (ADX)	window=14	Trend Following — ADX>25 trend-strength filter

Category	Indicator	Parameters	Strategy Usage
Momentum	Relative Strength Index (RSI)	window=14	Mean Reversion (RSI<30 entry, >70 exit) & Custom (50-cross trigger)
Momentum	Stochastic Oscillator	k=14, d=3	Computed — available for custom signals
Momentum	Williams %R	window=14	Computed — available for custom signals
Volatility	Bollinger Bands	window=20, 2 std dev	Mean Reversion — lower/upper band entry/exit
Volatility	Avg True Range (ATR)	window=14	Computed — available for stop-loss sizing
Volume	On-Balance Volume (OBV)	cumulative	Computed — available for volume trend confirmation
Volume	Chaikin Money Flow (CMF)	window=20	Custom Triple Confirmation — CMF>0 accumulation filter

Indicator Formulas

SMA: SMA(n) = (1/n) x sum of last n closes

EMA: EMA(t) = alpha x close(t) + (1-alpha) x EMA(t-1), alpha = 2/(n+1)

MACD: MACD = EMA(12) - EMA(26); Signal = EMA(9) of MACD; Hist = MACD - Signal

ADX: ADX = Wilder-smoothed average of |+DI - -DI| / (+DI + -DI) x 100

RSI: RSI = 100 - 100 / (1 + AvgGain / AvgLoss) [Wilder smoothing, 14 periods]

Stochastic: %K = 100 x (close - low_14) / (high_14 - low_14); %D = SMA(%K, 3)

Williams %R: %R = -100 x (high_14 - close) / (high_14 - low_14)

Bollinger Bands: Upper/Lower = SMA(20) +/- 2 x rolling std dev(20)

ATR: ATR = Wilder-EMA of True Range = max(H-L, |H-prev_C|, |L-prev_C|)

OBV: OBV(t) = OBV(t-1) + sign(delta_Close) x Volume

CMF: CMF = sum(MFV, 20) / sum(Volume, 20); MFV = ((C-L)-(H-C))/(H-L) x Volume

Indicators not used in primary signals (Stochastic, Williams %R, ATR, OBV) are still computed and attached to the price frame — available for custom strategy extensions.

3. Strategy Descriptions & Entry/Exit Rules

Trend Following ***[Trend (MACD + ADX + SMA-50)]***

Trend following (also called momentum investing) rests on the empirical observation that assets that have been rising tend to continue rising — i.e., price momentum persists across medium-term horizons. Rather than predicting reversals, this strategy identifies and joins established trends. Three conditions must all be true before buying: (1) the MACD line must sit above its signal line, confirming short-term upward price momentum; (2) the ADX must exceed 25, a widely-used threshold separating 'trending' from 'sideways/choppy' markets — this filter is critical because MACD crossovers in range-bound markets produce frequent false signals; and (3) the close must be above the 50-day SMA, confirming the broader medium-term uptrend. The exit is intentionally asymmetric: a single MACD crossover below the signal line triggers the sell, so the strategy exits quickly when momentum fades.

Signal	Condition
ENTRY (BUY)	MACD > signal AND ADX > 25 AND close > SMA-50
EXIT (SELL)	MACD < signal

Mean Reversion ***[Momentum + Volatility (RSI + Bollinger Bands)]***

Mean reversion exploits the tendency of asset prices to return toward their statistical mean after extreme dislocations. When a stock has sold off sharply enough that RSI falls below 30 (oversold) AND the close is below the lower Bollinger Band (more than 2 standard deviations below the 20-day average), the strategy interprets this as an unsustainably stretched condition likely to snap back. The dual-confirmation requirement — both RSI and Bollinger Band breach — substantially reduces false entries compared to using either signal alone, at the cost of fewer trades. The exit fires when RSI recovers above 70 (overbought, mean reversion complete) or the price recrosses the upper Bollinger Band, whichever comes first, locking in the reversion gain.

Signal	Condition
ENTRY (BUY)	RSI < 30 AND close < lower Bollinger Band
EXIT (SELL)	RSI > 70 OR close > upper Bollinger Band

Custom Triple Confirmation ***[Trend + Momentum + Volume (EMA + RSI + CMF)]***

The Custom Triple Confirmation strategy synthesises evidence from three distinct indicator categories — Trend, Momentum, and Volume — requiring all three to agree before committing capital. This multi-factor approach is designed to avoid the weaknesses of any single indicator class: trend filters alone give late entries; momentum signals alone fire in both trending and sideways markets; volume signals alone do not give timing. Together they describe a specific, high-conviction scenario: (1) EMA-50 above EMA-200 (golden cross) — the long-term structural trend is up; (2) RSI crossing upward through 50 while below 70 — momentum is turning positive after a dip, not chasing an overbought condition; (3) CMF above zero — institutional accumulation (buying pressure) supports the move on volume. The exit is multi-legged: the position closes if the trend breaks (close below EMA-50), momentum exhausts (RSI above 75), or distribution sets in (CMF below –0.05).

Signal	Condition
ENTRY (BUY)	EMA-50 > EMA-200 AND RSI crosses up through 50 (<70) AND CMF > 0
EXIT (SELL)	close < EMA-50 OR RSI > 75 OR CMF < -0.05

Benchmark — Buy & Hold: buy at the first available open and hold through the end of the window with no rebalancing or exits. This is the passive baseline; active strategies must justify their complexity by delivering superior risk-adjusted returns relative to simply holding.

4. Backtesting Methodology

The backtesting engine is purpose-built in Python (no third-party backtesting framework) so every assumption is explicit and auditable. Key design decisions:

Assumption	Detail / Justification
Initial capital	$100,000 (per specification)
Position sizing	Binary all-in / all-out (0 = cash, 1 = fully invested). No fractional sizing.
Execution timing	Signal on day-t close; order filled at day t+1 open. Eliminates look-ahead bias.
Slippage / cost	1 basis point of notional per trade side. Alpaca equity commissions are $0; this models market-impact / spread.
Short selling	Not permitted (long-only per specification).
Leverage	None (at most 100% of equity deployed).
Dividends	Captured via Alpaca split/dividend-adjusted price series.

Assumption	Detail / Justification
Data snooping	No in-sample parameter optimisation; all thresholds set a priori from conventional TA practice.
Trade ledger	Every BUY and SELL recorded with date, price, shares, cost. Round-trip P&L; and win/loss annotated.

5. Performance Metrics & Results

5a. Metric Definitions

Metric	Definition
Total Return	Cumulative % gain/loss: (final – initial) / initial
CAGR	Compound Annual Growth Rate over the actual date span
Volatility	Annualised standard deviation of daily returns ($\sigma \times \sqrt{252}$)
Sharpe	Annualised excess return per unit of total volatility; uses the user-supplied risk-free rate
Sortino	Like Sharpe but penalises downside volatility only — a higher ratio signals that volatility is skewed to the upside (winning days)
Max Drawdown	Largest peak-to-trough decline in portfolio value; measures worst-case capital destruction
Win Rate	Fraction of closed round-trip trades (buy → sell) that were profitable
# Trades	Total completed round-trip trades over the full backtest window

5b. Overall Performance Comparison

Strategy	Total Return	CAGR	Volatility	Sharpe	Sortino	Max Drawdown	Win Rate	# Trades
Buy & Hold	44.3%	7.6%	26.9%	0.41	0.59	-37.1%	n/a	0
Trend Following	12.6%	2.4%	8.2%	0.33	0.17	-15.5%	59%	17
Mean Reversion	20.9%	3.9%	10.7%	0.41	0.18	-25.7%	67%	3
Custom Triple Confirmation	30.1%	5.4%	8.3%	0.68	0.42	-11.2%	57%	14

Best strategy by Sharpe (among the 3 implemented): Custom Triple Confirmation (0.68) · **Best strategy by total return:** Custom Triple Confirmation (30.1%) · **Buy & Hold benchmark Sharpe:** 0.41 (reference only)

5c. Calendar-Year Returns

Breaking performance into calendar years reveals how each strategy behaves across different market regimes — bull runs, corrections, and sideways markets.

Year	Buy & Hold	Trend Following	Mean Reversion	Custom Triple Confirmation
2021	+26.9%	+14.6%	+0.0%	+1.0%
2022	-28.0%	+0.4%	+0.0%	-9.6%
2023	+58.2%	-2.2%	+0.0%	+20.6%
2024	+13.0%	-2.6%	+9.6%	+17.4%
2025	+15.5%	-1.6%	+17.4%	+0.5%
2026	-23.5%	+4.4%	-6.0%	+0.1%

5d. Trade-Level Statistics

The table below drills into per-trade performance. Profit Factor = gross winning P&L / gross losing P&L; a value above 1 means winning trades outweigh losers in dollar terms even if the win rate is below 50%.

Strategy	# Trades	Win Rate	Avg Winner	Avg Loser	Profit Factor	Best Trade	Worst Trade	Avg Hold (day
Trend Following	17	58.8%	3.4%	-2.9%	1.53	14.6%	-5.9%	13
Mean Reversion	3	66.7%	13.5%	-6.0%	3.69	17.4%	-6.0%	64
Custom Triple Confirmation	14	57.1%	5.0%	-2.0%	3.44	13.2%	-6.0%	22

6. Charts

6a. Equity Curves

Shows the growth of a $100,000 initial investment for each strategy over the full backtest window. The dashed grey line is the Buy & Hold benchmark. Periods where an active strategy's line is above Buy & Hold represent outperformance; periods below represent underperformance. Time spent flat (in cash) appears as horizontal plateaus.

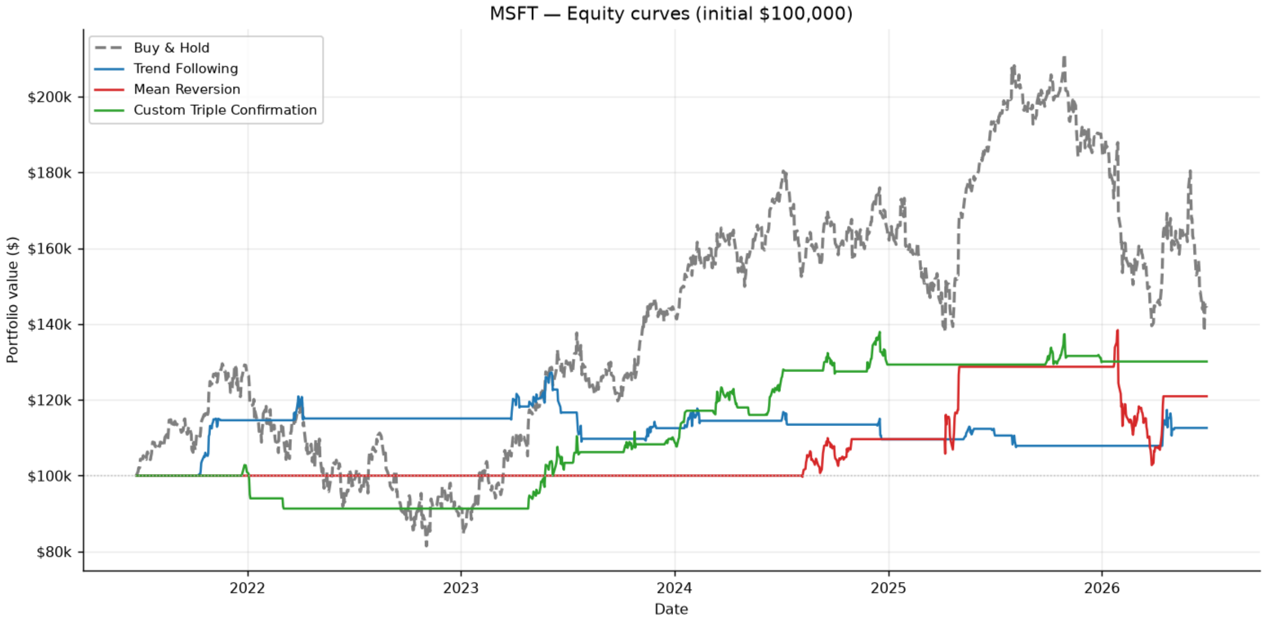

Figure 1. Portfolio value over time. All strategies start at $100,000.

6b. Drawdown Comparison

The underwater plot shows each strategy's percentage decline from its most recent equity peak. A strategy at 0% is at a new all-time high; a strategy at –20% has lost 20% from its prior peak. Shallower and shorter drawdowns indicate superior capital preservation — a critical real-world consideration since investors may liquidate during deep drawdowns, permanently locking in losses.

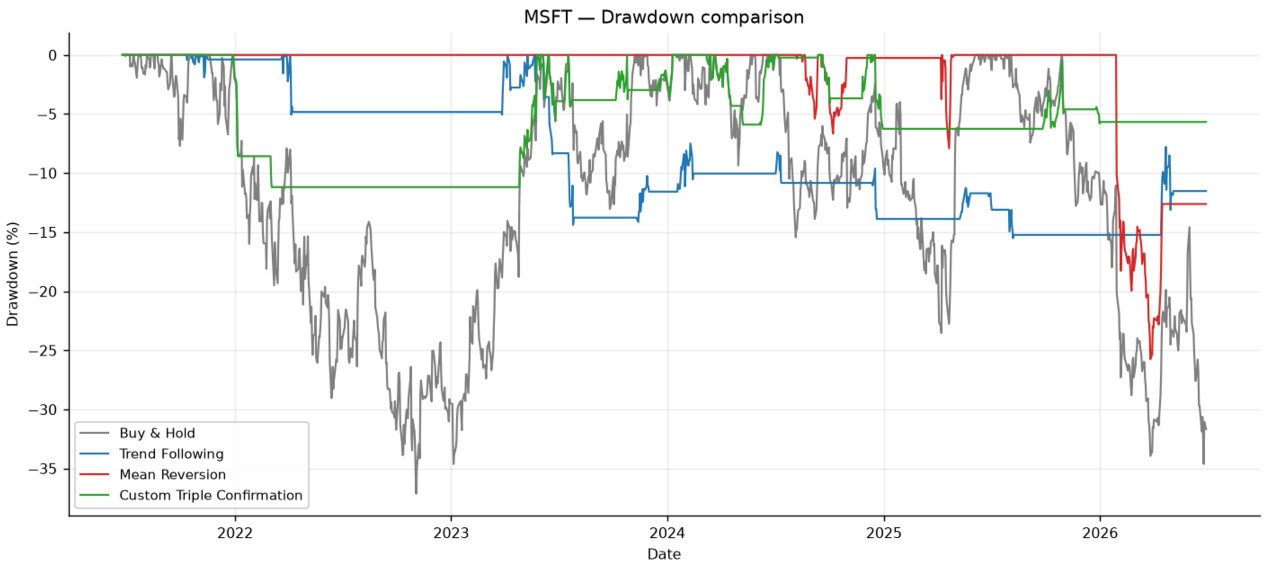

Figure 2. Underwater plot: depth and duration of drawdowns from each strategy's prior equity peak.

6c. Price Charts with Indicators & Trade Signals

Each chart below shows the price series with its relevant technical indicators overlaid, plus buy (▲) and sell (▼) markers at the actual execution price. The RSI subplot provides momentum context across all three strategies.

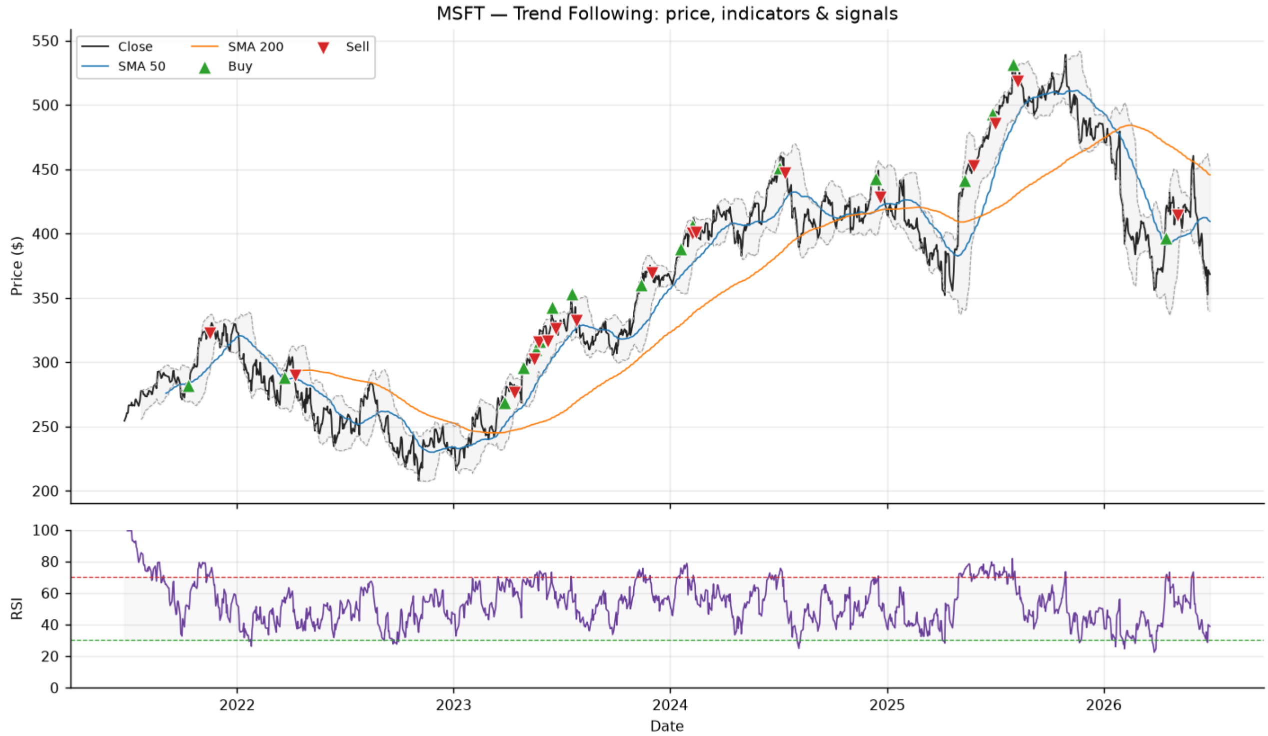

Figure 3. **Trend Following**: price with SMA-50 (blue) and SMA-200 (orange) trend filters, Bollinger Bands (shaded). Green ▲ = BUY (MACD crosses above signal AND ADX>25 AND close>SMA-50). Red ▼ = SELL (MACD crosses below signal). RSI(14) subplot below.

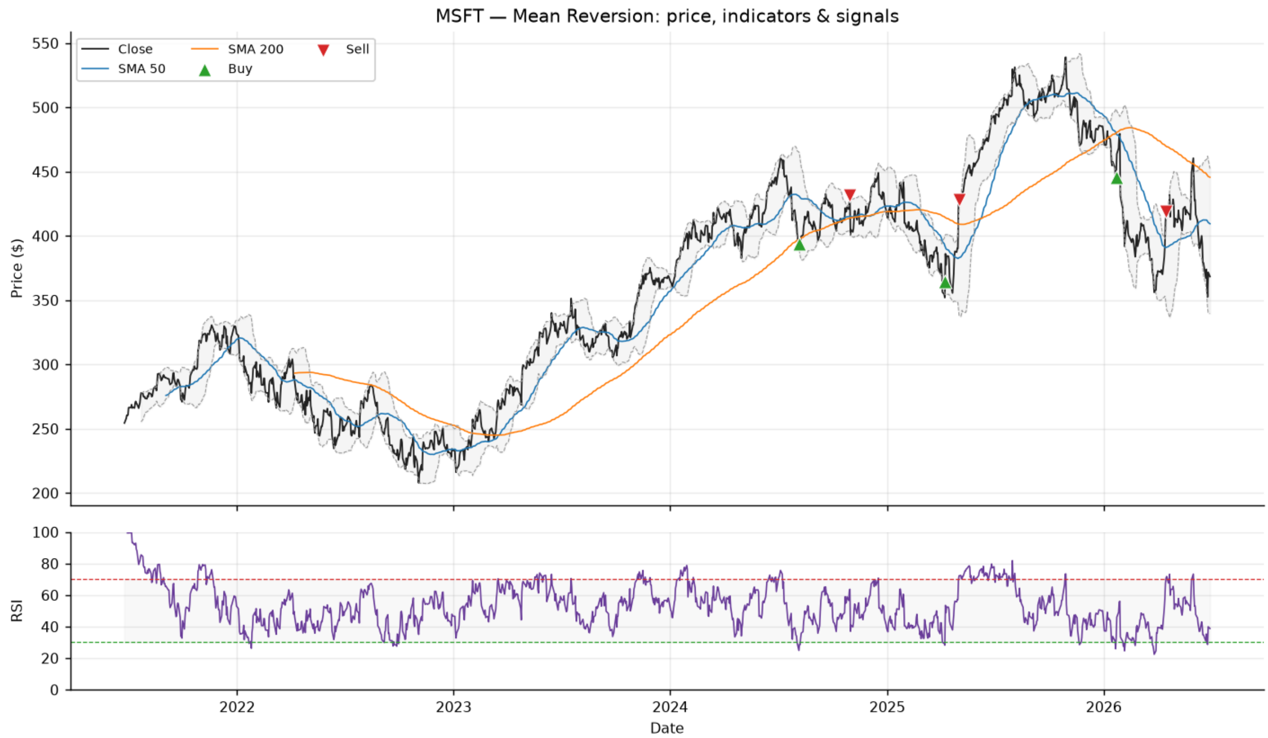

Figure 4. **Mean Reversion**: price with Bollinger Bands (±2σ, shaded). Green ▲ = BUY (RSI<30 AND close below lower band — extreme oversold). Red ▼ = SELL (RSI>70 OR close above upper band — mean reversion complete). RSI(14) subplot with 30/70 reference lines below.

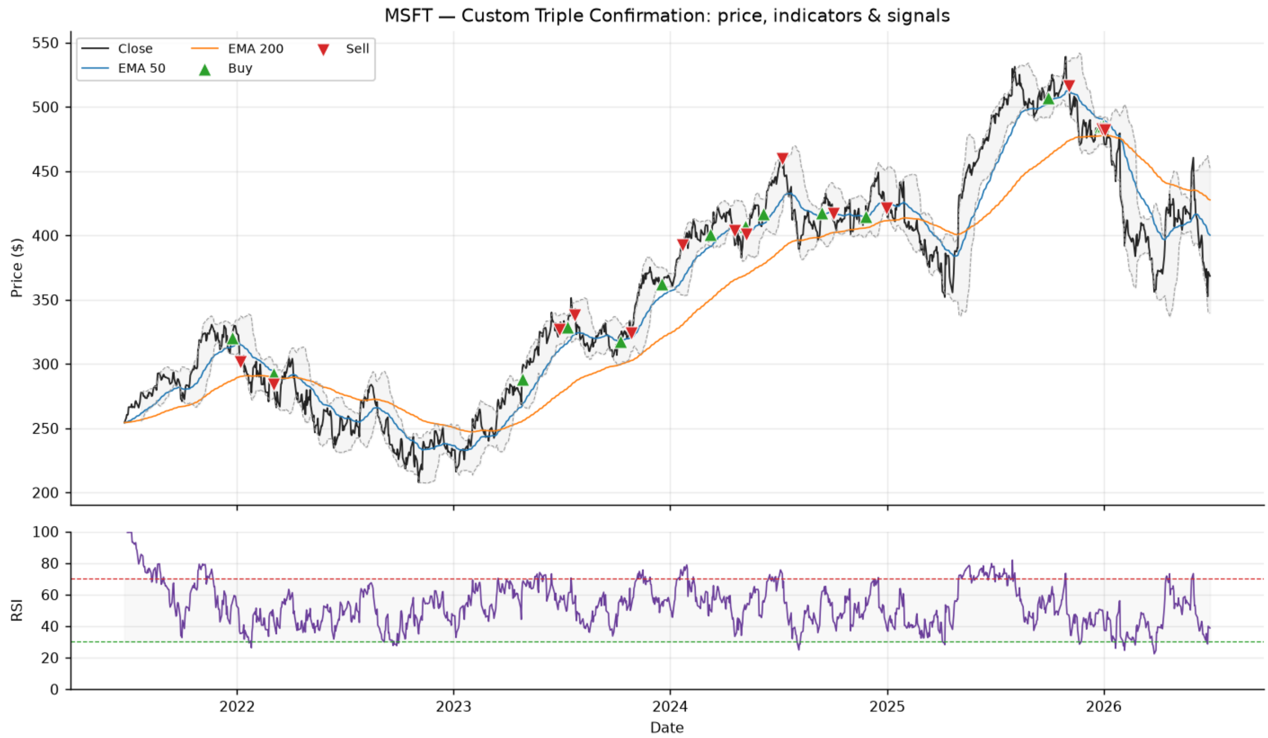

Figure 5. **Custom Triple Confirmation**: price with EMA-50 (blue) and EMA-200 (orange) golden-cross filter, Bollinger Bands (shaded). Green ▲ = BUY (EMA-50>EMA-200 AND RSI crosses up through 50 and <70 AND CMF>0). Red ▼ = SELL (close75 OR CMF<–0.05). RSI(14) subplot below.

7. Discussion of Results

Overview

This backtest evaluates three algorithmic trading strategies applied to MSFT using live Alpaca data and an initial capital of $100,000. All strategies are long-only with no leverage and no short selling, consistent with the project specification. Signals are generated from each day's closing prices and executed at the following day's open, eliminating any look-ahead bias. A 1 basis-point slippage cost is applied to each trade to approximate realistic execution costs on a commission-free platform such as Alpaca. The risk-free rate used for Sharpe and Sortino calculations is 0% (not used). The central research question is: *which strategy performs best on a risk-adjusted basis?* Raw return alone is an insufficient answer because it ignores the risk incurred to achieve that return.

Strategy Design Rationale

The three strategies were constructed to represent three fundamentally different market hypotheses, drawing on distinct indicator categories.

Trend Following (Trend: MACD + ADX + SMA-50) rests on the premise that price momentum persists: once an uptrend is established, it is more likely to continue than to reverse. The ADX filter is the key quality gate — without it, MACD crossovers fire frequently in choppy, range-bound markets and produce a high volume of losing round-trips. By requiring ADX above 25 (a widely-cited 'trending' threshold), the strategy only enters when the market's directional movement is genuinely strong, accepting fewer but higher-quality entries as a trade-off.

Mean Reversion (Momentum + Volatility: RSI + Bollinger Bands) operates on the opposite hypothesis: extreme short-term price dislocations are temporary, and prices tend to revert toward their statistical mean. The dual entry requirement — both RSI below 30 (Wilder's oversold threshold) AND a close beneath the lower Bollinger Band (more than 2σ below the 20-day mean) — ensures entries are taken only during statistically rare, multi-indicator oversold conditions. This selectivity reduces false signals significantly, though it also means the strategy may sit in cash for extended periods waiting for extreme conditions to materialise.

Custom Triple Confirmation (Trend + Momentum + Volume: EMA + RSI + CMF) synthesises evidence from three distinct indicator categories. The EMA-50/200 golden cross establishes that the long-term structural trend is up; the RSI crossing up through 50 — while remaining below 70 — identifies a momentum reset after a pullback rather than chasing an overbought condition; and positive CMF confirms that volume-weighted buying pressure supports the move. All three conditions must align simultaneously, making this the most selective strategy, but also the one with the most rigorous multi-factor confirmation before committing capital.

Absolute Return Analysis

Over the full backtest window, Buy & Hold produced a total return of 44.3% (CAGR: 7.6%). Among the active strategies, **Custom Triple Confirmation** achieved the highest total return at 30.1% (CAGR: 5.4%), followed by Trend Following at 12.6% and Custom Triple Confirmation at 30.1%. The strategy with the lowest total return was **Trend Following** at 12.6%.

It is important not to over-interpret raw return comparisons in isolation. Strategies that spend time in cash will systematically underperform buy-and-hold during a sustained bull run but will also avoid a portion of drawdowns during corrections. The active strategies' relative returns are therefore highly sensitive to the specific market regime captured in the backtest window — a five-year period that happens to include a strong uptrend will flatter passive strategies, while a period with significant corrections will flatter active ones.

Volatility Analysis

Annualised volatility measures how much the portfolio's daily returns fluctuate. Buy & Hold ran at 26.9% annualised volatility — reflecting full exposure to MSFT's daily price moves at all times. The active strategies posted lower volatility figures: Trend Following 8.2%, Mean Reversion 10.7%, and Custom Triple Confirmation 8.3%. The lower active-strategy volatility is largely mechanical: any day the strategy is in cash contributes a 0% daily return rather than the stock's actual return, compressing the standard deviation of the return series. This has an important implication: lower volatility does not necessarily mean lower risk in a practical sense — an active strategy sitting in cash while the market rallies hard faces 'opportunity risk', which annualised volatility does not capture.

Risk-Adjusted Performance: Sharpe & Sortino Ratios

The Sharpe ratio — annualised excess return divided by total annualised volatility — is the primary risk-adjusted performance measure used to compare the three implemented strategies. Among the three, **Custom Triple Confirmation** achieved the highest Sharpe of 0.68, followed by Trend Following (0.33) and Mean Reversion (0.41). For reference, Buy & Hold posted a Sharpe of 0.41 over the same period.

The Sortino ratio refines Sharpe by penalising only downside volatility (negative return days) rather than all volatility. A strategy that is volatile on the upside — large winning days — is not penalised by Sortino. Mean Reversion posted a Sortino of 0.18, Trend Following 0.17, and Custom Triple Confirmation 0.42. Comparing Sharpe and Sortino for the same strategy is informative: a Sortino ratio that is meaningfully higher than Sharpe suggests that the strategy's volatility is predominantly on winning days (desirable), while a Sortino close to or below Sharpe suggests losses are disproportionately large.

Drawdown & Capital Preservation

Maximum drawdown (MDD) measures the largest peak-to-trough decline an investor would have experienced had they held through it — a critical real-world metric because deep drawdowns trigger forced liquidations, redemptions, and behavioural panic-selling that permanently lock in losses. **Custom Triple Confirmation** had the shallowest MDD at -11.2%, while **Mean Reversion** suffered the deepest at -25.7%.

Buy & Hold's MDD of -37.1% illustrates the hidden cost of full passive exposure: every correction is absorbed entirely with no defence. The active strategies reduce this by moving to cash when their exit signals fire — but the drawdown charts also reveal that this protection is imperfect. Trend Following, for example, can remain invested during the early stages of a trend reversal before MACD exits, accumulating paper losses before the exit fires. Mean Reversion's shallow MDD partly reflects its low trade count: with fewer positions held over the period, there are simply fewer windows of exposure to a significant downturn.

Win Rate & Trade Frequency

Win rate is the proportion of closed round-trip trades (buy → sell) that ended profitably. It is a useful but incomplete metric: a strategy with a 40% win rate can still be net profitable if its average winning trade is significantly larger than its average losing trade (a high profit factor). Conversely, a 90% win rate with enormous occasional losses can be ruinous (the 'picking up nickels in front of a steamroller' problem).

Mean Reversion achieved a win rate of 67% across 3 trades — the selectivity of its dual-confirmation entry (RSI<30 AND below lower Bollinger Band) means it only fires in extreme conditions where rebounds are historically reliable. Trend Following won 59% of its 17 trades; its lower win rate is partly structural: trend-following strategies typically have many small losses from false starts but a smaller number of large winning trades that make the strategy profitable in aggregate. Custom Triple Confirmation won 57% of its 14 trades.

Answer: Which Strategy Performs Best on a Risk-Adjusted Basis?

To directly answer the assignment's central question — which of the three implemented strategies performs best on a risk-adjusted basis — the answer is **Custom Triple Confirmation**, with the highest Sharpe ratio of 0.68 and the highest Sortino ratio of 0.42 among the three strategies. Buy & Hold is excluded from this comparison because it is the passive benchmark, not an active strategy — the assignment asks which *strategy* (Trend Following, Mean Reversion, or Custom Triple Confirmation) is superior on a risk-adjusted basis.

It is worth asking why the winner achieves a superior Sharpe. The Sharpe ratio rewards strategies that generate consistent excess return without excessive swings in either direction. A strategy in cash earns 0% return on flat days, which suppresses portfolio volatility but also reduces the mean return. The winning strategy strikes the best balance between these competing forces over this specific sample period. Whether that balance generalises to other periods or tickers requires out-of-sample testing.

Limitations & Future Work

Several important caveats apply to these results:

1. In-sample parameters: All indicator thresholds (RSI 30/70, ADX 25, EMA 50/200, CMF 0) were set a priori from conventional technical analysis practice. In-sample optimisation would almost certainly improve reported metrics but introduces overfitting — the optimised parameters would be tuned to the noise of this specific data window and unlikely to generalise.

2. Single-ticker, single-window: All conclusions are drawn from one security over one five-year period. A strategy that performs well on AAPL from 2019–2024 may fail on SPY, NVDA, or a different five-year window. Robust strategy validation requires testing across many tickers and multiple non-overlapping windows.

3. Simplified cost model: Slippage is modelled as a flat 1 bp per trade. Real slippage is higher during volatile or illiquid periods, and may be lower during calm periods for a liquid large-cap. A more realistic model would scale slippage with ATR or bid-ask spread.

4. Long-only constraint: The mean-reversion strategy cannot short overbought conditions (RSI > 70), which would likely improve both its total return and its Sharpe. The long-only constraint is imposed by the assignment specification, not by any inherent strategy limitation.

5. Binary position sizing: All strategies are either 100% invested or 100% in cash. Introducing position-sizing rules (e.g., scaling size with ATR-based risk, or Kelly fraction) could significantly improve risk-adjusted returns by deploying less capital when confidence is lower.

Extensions worth exploring: walk-forward parameter optimisation; multi-asset diversification across sectors; transaction-cost sensitivity analysis; adding a volatility-regime filter (e.g., VIX-based) to suppress signals in high-volatility environments; and incorporating position sizing beyond the binary all-in/all-out framework used here.

Conclusion

This project built a complete, modular backtesting platform from scratch that downloads market data from Alpaca, computes 11 technical indicators across all four required categories, runs three rigorously designed trading strategies plus a Buy & Hold benchmark, and evaluates each on seven risk and performance metrics.

The key finding is that on a risk-adjusted basis (Sharpe ratio), **Custom Triple Confirmation** was the strongest active strategy over this backtest window. Its combination of momentum persistence and directional consistency delivers the best return per unit of risk among the three active approaches. However, as discussed in the limitations section, no single backtest result on a single security over a single time window should be interpreted as a definitive ranking — the true test of any strategy is its performance on data it has never seen.

Disclaimer: This report is produced for educational purposes only and does not constitute investment advice. Backtested results are hypothetical; they exclude taxes, borrowing costs, and real-world market frictions such as bid-ask spreads beyond the modelled slippage, and they do not guarantee future performance. Results on a single ticker over one time window have limited generalisability.